



FDR4VGT Atmospheric Correction

Algorithm Theoretical Basis Document (ATBD)

Référence :	FDR4VGT-AC-ATBD
Date :	2026-07-17
Version :	1.1
Numéro de contrat :	FDR4VGT
Nom du client :	C3S

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0.1 Version History

Version	Date	Author(s)	Description
v1.0	2024-01-XX	HYGEOS	Initial ATBD for FDR4VGT atmospheric correction
v1.1	2025-07-17	HYGEOS	Implementation notes, performance optimizations, float32 fix, known limitations
v1.1	2026-07-17	HYGEOS	Addendum: Appendix E with SMACCL OpenCL kernel and wrapper API contract

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0.2 Acronyms

Acronym	Definition
AC	Atmospheric Correction
AERONET	AErosol RObotic NETwork
AOD	Aerosol Optical Depth
AMIP	Atmospheric Model Intercomparison Project
ARTDECO	Atmospheric Radiative Transfer Data Centre
BRDF	Bidirectional Reflectance Distribution Function
CAMS	Copernicus Atmosphere Monitoring Service
C3S	Copernicus Climate Change Service
CF	Climate and Forecast (metadata conventions)
CTM	Chemical Transport Model
DEM	Digital Elevation Model
DTOA-r	Error on TOA reflectance
FDR	Full Dynamic Range
FIDUCEO	Fiducial Data for Earth Observation
GPU	Graphics Processing Unit

Acronym	Definition
HITRAN	High-Resolution Transmission Molecular Database
L1C	Level 1C (TOA reflectance product)
L2A	Level 2A (TOC reflectance product)
LUT	Look-Up Table
MERRA-2	Modern-Era Retrospective analysis for Research and Applications, Version 2
MLP	Multi-Layer Perceptron
NA	Not Applicable
ONNX	Open Neural Network Exchange
OPAC	Optical Properties of Aerosols and Clouds
PDF	Probability Density Function
PSF	Point Spread Function
RAA	Relative Azimuth Angle
RMSE	Root Mean Square Error
RTM	Radiative Transfer Model
RSRF	Relative Spectral Response Function
SMAC	Simplified Method for Atmospheric Correction
SMAC-CL	SMAC implementation in OpenCL
SZA	Solar Zenith Angle
TOA	Top of Atmosphere
TOC	Top of Canopy
UHO	Total Column Water Vapor
UO	Total Column Ozone
VGT	Vegetation Monitoring Tool (SPOT)
VZA	View Zenith Angle
VAA	View Azimuth Angle

1. Introduction

This Algorithm Theoretical Basis Document (ATBD) describes the atmospheric correction (AC) algorithm for the Full Dynamic Range (FDR) Vegetation Monitoring Tool (VGT) product, including both SPOT-VGT and Proba-V sensors. The algorithm implements the Simplified Method for Atmospheric Correction (SMAC) with rigorous uncertainty propagation following the FIDUCEO (Fiducial Data for Earth Observation) methodology.

Version 1.1 updates:

- Documents all performance optimizations implemented in the production codebase (tile processing, Zarr caching, local I/O, COW shared state)
- Specifies float32 data type requirement for SLP/T10M fields to prevent float16 overflow in pressure computation
- Updates DEM interpolation note (linear interpolation improvement over nearest-neighbor)
- Documents Köppen-Geiger regionalization as a planned feature
- Confirms adjacency effects as a known limitation
- Adds implementation-specific details (tile grid, ONNX model inputs, quality flag bits, output NetCDF variables)
- Adds the SMACCL OpenCL kernel and Python wrapper API contract (Appendix E)

The FDR4VGT AC algorithm provides:

- **Top-of-Canopy (TOC) directional reflectance** for 4 spectral bands (Blue, Red, NIR, SWIR)
- **Complete uncertainty budget** with propagation of all major uncertainty sources
- **Quality flags** for cloud contamination, high AOD, high solar zenith angle, and AOD gradients
- **Jacobian matrices** for sensitivity analysis of TOC reflectance to input parameters

2. Input and Auxiliary Data

2.1 Level-1C Input

The input data is the Level-1C (L1C) FDR product, which provides TOA reflectance at the sensor spectral bands with associated geometry and quality information.

Supported sensors:

Table 1: *Supported sensors and bands*

Sensor	Bands	Spatial Resolution
SPOT-VGT	B1 (Blue), B2 (Red), B3 (NIR), B4 (SWIR)	~1 km (nadir)
Proba-V	B1 (Blue), B2 (Red), B3 (NIR), B4 (SWIR)	~1 km (nadir)

Required input variables:

Table 2: *Required L1C input variables*

Variable	Description	Dimensions
TOA	TOA reflectance	(bands, y, x)
DTOA-r	TOA reflectance error	(bands, y, x)
SZA	Solar Zenith Angle	(y, x)
SAA	Solar Azimuth Angle	(y, x)
VZA	View Zenith Angle (VNIR)	(y, x)
VAA	View Azimuth Angle	(y, x)
VZA_IR	View Zenith Angle (SWIR, Proba-V)	(y, x)
VAA_IR	View Azimuth Angle (SWIR, Proba-V)	(y, x)
clm	Cloud mask	(y, x)
lat	Latitude	(y, x)
lon	Longitude	(y, x)

Notes:

- Proba-V has a dual-FOV sensor (VNIR and SWIR with different viewing geometries), hence separate VZA_IR/VAA_IR for the SWIR band
- Cloud mask (clm) values: 0 = clear, non-zero = cloud-contaminated
- Sensor readers use `core.interpolate` for resampling (generic xarray-based interpolation)

2.2 Digital Elevation Model

The Digital Elevation Model (DEM) provides terrain elevation for surface pressure correction and terrain effect estimation.

Source: GTOPO30 (90m global resolution)

Fields:

Table 3: *DEM fields*

Field	Description	Units
elev	Terrain elevation	meters
Delev	Elevation difference (for uncertainty)	meters

Interpolation: Linear interpolation to satellite grid

Implementation Note (v1.1): The implementation uses linear interpolation (via `scipy.interpolate.RegularGridInterpolator`) rather than the nearest-neighbor method specified in ATBD v1.0. This is a **reasonable improvement** — linear interpolation provides smoother elevation transitions at tile boundaries. The impact is negligible for the 90m GTOPO30 resolution vs ~1km VGT pixels.

DEM Cropping (v1.1): The global GTOPO30 DEM is cropped to the product's lat/lon extent before loading, reducing memory from ~1.9 GB to ~0.2 GB for typical narrow longitude ranges.

2.3 MERRA-2 Auxiliary Data

MERRA-2 (Modern-Era Retrospective analysis for Research and Applications, Version 2) provides atmospheric state variables for atmospheric correction.

Source: MERRA-2 instantaneous 2D aerosol and meteorological fields

Required fields:

Table 4: *MERRA-2 fields and unit conversions*

Variable	Description	MERRA-2 Unit	SMAC Unit	Conversion
T03	Total ozone column	Dobson	cm.atm	E1e-3
TQV	Total column water vapor	kg.mš	g.cmš	E1e-1
SLP	Sea-level pressure	Pa	hPa	E1e-2
T10M	Temperature at 10m	K	K	—
TOTEXTTAU	Total AOD at 550nm	—	—	—

Aerosol species fractions:

Table 5: *Aerosol species fractions*

Variable	Description
SU_FRAC	Sulfate fraction
DU_FRAC	Dust fraction
OC_FRAC	Organic carbon fraction
SS_FRAC	Sea salt fraction
BC_FRAC	Black carbon fraction

Interpolation: `scipy.interpolate.RegularGridInterpolator` with `method='linear'` (mathematically equivalent to bilinear interpolation for regular grids). Temporal interpolation includes time as a third dimension (3D interpolation: time, lat, lon).

Implementation Note (v1.1): The implementation pre-interpolates MERRA-2 fields to the full satellite grid once and stores them in a local Zarr cache (`precompute_ancillary()`). Tiles are then sliced from the local cache. This is numerically equivalent to per-tile interpolation but provides significant I/O performance improvement (see Section 11.1).

Critical Data Type Requirement (v1.1): SLP and T10M fields **must be stored as float32** (not float16) in the ancillary cache. The surface pressure computation `Ps()` in `smaccl.c3s_lib` evaluates $\log(R \cdot T)$ where $R \cdot T = 287.058 \times 290 \times 83246$, which exceeds the float16 maximum value (65504), causing overflow and NaN propagation. This is enforced via `_MERRA_CACHE_FLOAT32_VARS = {'SLP', 'T10M'}` in the implementation.

2.4 AMIP Ensemble

The AMIP (Atmospheric Model Intercomparison Project) ensemble provides 10 members of monthly aerosol climatology for uncertainty estimation through ensemble spread.

Source: MERRA-2 AMIP-10 ensemble reanalysis (`m2amip01.tavgM_2d_aer_Nx*.nc4`)

Fields: Same aerosol fields as MERRA-2, per ensemble member

Processing:

- 10 ensemble members are loaded monthly
- Monthly climatology is computed (no temporal interpolation)
- For years 2018-2020, the 2017 climatology is used as fallback

- Each ensemble member provides a complete set of aerosol species for model selection

Aerosol model selection from AMIP: For each ensemble member j , the closest aerosol model `iaero_AMIPj` is selected using the same Euclidean distance method as for the primary MERRA-2 data (see Section 5.2.6).

2.5 SMAC Coefficients

SMAC coefficients are pre-computed lookup tables that parameterize the radiative transfer model for 148 aerosol models.

Format: Pre-computed `.npy` or `.nc` files

Structure:

Table 6: *SMAC coefficient table structure*

Dimension	Size	Description
Aerosol models	148	Predefined aerosol mixtures (see Appendix B)
Bands	4	VIS-NIR (B1-B3) + SWIR (B4)
Coefficients	Variable	SMAC polynomial coefficients

Coefficient indices: Stored in `ca_ind` dictionary mapping names (`a0taup`, `a1taup`, `taur`, etc.) to array indices.

Loading: `read_smac_coefficients()` supports both `.npy` (structured array) and `.nc` (NetCDF) formats with shape homogenization for broadcast compatibility.

2.6 C3S Surface Albedo / BRDF

The C3S (Copernicus Climate Change Service) Multi-Sensor Surface Albedo product provides BRDF coefficients for non-Lambertian surface correction.

Source: C3S L3 BRDF product (RossThick-LiSparse-R model)

Fields:

Table 7: *C3S BRDF fields*

Field	Description
<code>k0</code>	Isotropic albedo coefficient

Field	Description
k1	Geometric kernel coefficient
k2	Volumetric kernel coefficient

Normalization: BRDF shape parameters are normalized as:

$$k_1^p = \frac{k_1}{k_0}; \quad k_2^p = \frac{k_2}{k_0}$$

Usage: In the baseline run, the surface is assumed Lambertian ($k1p = k2p = 0$). A second run with BRDF coefficients from C3S allows estimation of the BRDF effect on TOC reflectance.

Interpolation: Bilinear interpolation to satellite grid via `load_brdf()`.

3. Atmospheric Correction Algorithm

3.1 SMAC Overview

The Simplified Method for Atmospheric Correction (SMAC) approximates the radiative transfer equation with fast polynomial fits pre-computed from a dense sampling of exact RTM (6S/ARTDECO) simulations.

Core equations:

The TOA reflectance is modeled as:

$$R_{toa} = T_{dir} \cdot R_{toc} \cdot e^{-\frac{\tau}{\mu_s}} + R_a + R_{as} \quad (1)$$

Where:

- T_{dir} : Direct atmospheric transmission
- R_{toc} : TOC reflectance (unknown to be solved)
- τ : Total atmospheric optical depth (aerosol + Rayleigh)
- $\mu_s = \cos(\theta_s)$: Cosine of solar zenith angle
- R_a : Atmospheric reflectance (Rayleigh scattering only)
- R_{as} : Aerosol scattering reflectance

The atmospheric correction inverts this equation:

$$R_{toc} = \frac{R_{toa} - R_a - R_{as}}{T_{dir} \cdot e^{-\frac{\tau}{\mu_s}}} \quad (2)$$

The atmospheric correction factor (η) is:

$$\eta = \frac{1}{T_{dir} + s_{atm} \cdot R_{toc}} \quad (3)$$

Where s_{atm} accounts for multiple scattering between atmosphere and surface.

Gaseous absorption: SMAC coefficients include pre-fitted gaseous transmission terms for O and HO:

$$T_{gas} = \exp(-a \cdot (m \cdot U_{gas})^n) \quad (4)$$

Where m is the air mass, U_{gas} is the gas column amount, and a , n are SMAC coefficients fitted from line-by-line RT calculations (HITRAN database).

3.2 Sensitivities and Jacobians

The uncertainty propagation requires the sensitivity of R_{toc} to each input parameter (Jacobian matrices).

Analytical Jacobians:

Table 8: *Analytical Jacobians*

Parameter	Jacobian	Method
R_{toa}	$J_{R_{toa}} = \partial R_{toc} / \partial R_{toa}$	Analytical (Eq. 6-7)
U_{O_3}	$J_{U_{O_3}} = \partial R_{toc} / \partial U_{O_3}$	Analytical (Eq. 8-13)
U_{H_2O}	$J_{U_{H_2O}} = \partial R_{toc} / \partial U_{H_2O}$	Analytical (Eq. 8-13)

Finite-difference Jacobians:

Table 9: *Finite-difference Jacobians*

Parameter	Jacobian	Perturbation
P_s	$J_{P_s} = \partial R_{toc} / \partial P_s$	$\delta P_s = 10 \text{ hPa}$
τ_a^{550}	$J_{\tau_a} = \partial R_{toc} / \partial \tau_a$	$\delta \tau = 0.1 \cdot \tau$

All Jacobians are computed in the SMAC-CL OpenCL kernel (pixel-parallel execution).

3.3 Atmospheric Transmissions

The total atmospheric transmission is decomposed into:

$$T = T_{dir} + T_{dif} \quad (5)$$

Where:

- T_{dir} : Direct (beam) transmission
- T_{dif} : Diffuse (scattered) transmission

Optical depths:

Aerosol optical depth (from SMAC coefficients):

$$\tau_a = a_0 + a_1 \cdot AOD_{550} \quad (6)$$

Rayleigh optical depth (from SMAC coefficients):

$$\tau_r = \tau_{r,coef} \cdot P \quad (7)$$

Where P is the equivalent pressure.

4. Quality Flags

Quality flags are constructed as a bitfield (int16) with the following bits:

Table 10: *Quality flag bit definitions*

Bit	Name	Condition	Description
0	Cloud	<code>clm != 0</code>	Cloud-contaminated pixel
1	High AOD	<code>TOTEXTTAU >= 0.6</code>	High aerosol loading
2	High SZA	<code>SZA > 80°</code>	Large solar zenith angle
3	AOD Gradient	<code>aod_grad > threshold</code>	High AOD spatial gradient

Bit	Name	Condition	Description
4	AC Failure	rTOC is NaN/invalid	Atmospheric correction algorithm failure
5	Missing Aux	MERRA-2 field is NaN	Missing auxiliary data
6	Out of LUT	Coefficient bounds exceeded	Out of SMAC lookup table range

Flag construction: `build_flag()` in `funcs.py` implements all 7 flag bits with configurable thresholds.

AOD gradient: Computed as `aod_grad = sqrt(gradient_yš + gradient_xš)` using `da.gradient()` (dask gradient).

5. Uncertainty Analysis

5.1 Primary Uncertainty Equation

Following the FIDUCEO methodology, the total uncertainty on R_{toc} is computed by propagating all input uncertainties through the measurement equation:

$$u^2(R_{toc}) = \sum_{i=1}^7 J_i^2 \cdot u^2(x_i) + u_{ensemble}^2 + u_{BRDF}^2 + u_{RTM}^2 + u^2(0) \quad (19)$$

Where the 7 propagated effects are:

Table 11: *The 7 propagated uncertainty effects*

Effect	Parameter	Jacobian	Uncertainty
1	R_{toa}	$J_{R_{toa}}$	$u(R_{toa})$ from L1C
2	U_{H2O}	$J_{U_{H2O}}$	$u(U_{H2O})$ from MERRA-2
3	U_{O3}	$J_{U_{O3}}$	$u(U_{O3})$ from MERRA-2
4	P_s	J_{P_s}	$u(P_s)$ from DEM + meteo
5	τ_a^{550}	J_{τ_a}	$u(\tau_a)$ from validation
6	Aerosol model	—	Ensemble spread
7	RTM approximation	—	BRDF + terrain + SMAC fit

Implementation: `compute_urtoe()` in `process.py` implements Eq. 19 with all 9 terms,

returning both total uncertainty and individual component absolute values for diagnostics.

5.2 Uncertainty Effects

5.2.1 Effect 1: TOA Reflectance Uncertainty

The TOA reflectance uncertainty is derived from the L1C product error components:

$$\sigma(R_{toa}) = \sqrt{u_{random}^2 + u_{structured}^2 + u_{systematic}^2} \quad (20)$$

Where `UNC_RANDOM`, `UNC_STRUCTURED`, and `UNC_SYSTEMATIC` are band-dependent uncertainty components from the L1C DTOA-r product.

5.2.2 Effect 2: Ozone Uncertainty

Table 12: *Ozone uncertainty parameters*

Parameter	Value	Source
Relative uncertainty	6% of UO	Wargan et al. (2017), Davis et al. (2017)
Spatial correlation	~500 km (synoptic)	Meteorological scale
Temporal correlation	~3 days	Ozone lifetime

5.2.3 Effect 3: Water Vapor Uncertainty

Table 13: *Water vapor uncertainty parameters*

Parameter	Value	Source
Relative uncertainty	20% of UHO	Conservative estimate
Spatial correlation	~200 km (convective)	Meteorological scale
Temporal correlation	~1 day	Water vapor lifetime

5.2.4 Effect 4: Surface Pressure Uncertainty

Surface pressure is computed from MERRA-2 sea-level pressure corrected for terrain elevation:

$$P_s = P_0 \cdot \exp\left(-\frac{g \cdot \Delta h}{R \cdot T}\right) \quad (21)$$

Where:

- P_0 : Sea-level pressure from MERRA-2 SLP
- $g = 9.80665$ m/s²: Gravitational acceleration
- $R = 287.058$ J/(kg·K): Gas constant for dry air
- $\Delta h = h_{pixel} - h_{MERRA}$: Elevation difference
- $T = T_{10M}$: Temperature at 10m from MERRA-2

The pressure uncertainty combines meteorological and elevation components:

$$u(P_s) = \frac{\sqrt{(\Delta_{P_s}^\lambda)^2 + (\Delta_{P_s}^z)^2}}{2} \quad (22)$$

Where $\Delta_{P_s}^\lambda \approx 1$ hPa (meteorological model uncertainty) and $\Delta_{P_s}^z$ from DEM elevation uncertainty.

Critical Implementation Note (v1.1): The computation of P_s involves the term $\log(R \cdot T)$ in the `smacc1.c3s_lib.Ps()` function. For typical temperatures $T \approx 290$ K, the product $R \cdot T = 287.058 \times 290 \approx 83246$ exceeds the float16 maximum value (65504), causing overflow and NaN propagation through all downstream computations. **Therefore, SLP and T10M fields must be stored and processed as float32.** This is enforced in the implementation via `_MERRA_CACHE_FLOAT32_VARS = {'SLP', 'T10M'}`.

5.2.5 Effect 5: AOD Uncertainty

The AOD uncertainty model is based on comprehensive MERRA-2 validation against AERONET (Gueymard & Yang, 2020):

$$\sigma(\tau_a^{550}) = \max(\sigma_{base}, \sigma_{relative} \times \tau_a^{550}) \quad (23)$$

Where:

- $\sigma_{base} = 0.03$ (clean atmosphere baseline)
- $\sigma_{relative} = 0.15$ (15% relative uncertainty)

Regional multipliers (planned feature):

Table 14: Köppen-Geiger regional AOD uncertainty multipliers (planned)

Region Type	Multiplier	Example Locations
Oceanic/Polar	0.7E	Open oceans, Arctic
North America/Europe	1.0E	Baseline reference
Dust-dominated	1.5E	Sahara, Arabian Peninsula
Biomass burning	1.4E	Amazon, Sub-Saharan Africa
High pollution	1.3E	East Asia, South Asia

Implementation Note (v1.1): Köppen-Geiger climate classification multipliers are documented as a **planned feature** for a future update. The current implementation uses the global model (Eq. 23) without regional multipliers.

5.2.6 Effect 6: Aerosol Model Uncertainty

The aerosol model uncertainty is quantified using the AMIP 10-member ensemble spread:

$$u_{ensemble}(R_{toc}) = \text{std}(R_{toc,j}), j = 1..10 \quad (24)$$

For each pixel:

1. Select 10 preferred aerosol models from 148 options (based on MERRA-2 species fractions matching AMIP ensemble)
2. Run SMAC with each model $R_{toc,1}...R_{toc,10}$
3. $u_{ensemble} = \text{standard_deviation}(R_{toc,j})$

5.2.7 Effect 7: RTM Approximation Uncertainty

Three sources of RT approximation errors:

7a. Lambertian Surface Assumption (BRDF Effect) The BRDF effect is estimated by comparing Lambertian vs BRDF-corrected runs:

$$u_{BRDF}(R_{toc}) = R_{toc}(k_{1p} = 0, k_{2p} = 0) - R_{toc}(k_{1p} \neq 0, k_{2p} \neq 0) \quad (25)$$

Significance: Moderate (NIR, high AOD). Magnitude: ~1% for most cases, up to 5% for strong BRDF + high AOD.

7b. Terrain Slope Effects The CESBIO terrain model computes the terrain-induced relative effect:

$$slope_{eff} = \frac{T}{T_{dir} \cdot \frac{\mu_i}{\mu_s} + T_{dif} \cdot f_{sky} + T \cdot f_{ground} \cdot \rho_{env}} \quad (26)$$

$$u_{terrain}(R_{toc}) = R_{toc} \cdot (slope_{eff} - 1) \quad (27)$$

Where:

- $\mu_s = \cos(\theta_s)$: Cosine of solar zenith angle
- $\mu = \cos(s)$: Cosine of terrain slope
- $\mu_i = \mu_s \cdot \mu + \sqrt{1 - \mu_s^2} \cdot \sqrt{1 - \mu^2} \cdot \cos(\phi_s - \phi_a)$: Effective cosine on tilted surface
- $f_{sky} = (1 + \mu)/2$: Sky view factor
- $f_{ground} = (1 - \mu)/2$: Ground view factor
- ρ_{env} : Environment reflectance (assumed equal to pixel reflectance)

Significance: Moderate to High in mountainous regions. Magnitude: Several %, up to 40% in extreme terrain.

7c. SMAC Polynomial Fitting Errors The SMAC fit uncertainty is estimated using an ONNX (Open Neural Network Exchange) model:

$$u_{RTM_fit}(R_{toc}) = f_{ONNX}(SZA, VZA, RAA, AOD, P_s, R_{toc}, iaer, wavelength) \quad (28)$$

ONNX model inputs:

Table 15: *ONNX RTM fit uncertainty model inputs*

Input	Description	Units
pression	Surface pressure	hPa
iaer	Aerosol model index	0-147
raa	Relative azimuth angle	degrees (0-180)
rtoc	Surface reflectance	—
sza	Solar zenith angle	degrees
tauaer	AOD at 550nm	—
vza	View zenith angle	degrees

Input	Description	Units
wvl	Wavelength	nm

Mean square error of the ONNX model: ~ 0.0097 on predicted RTM uncertainty.

5.3 Second-Order Effects

Effects considered minor and not explicitly propagated:

Table 16: *Second-order uncertainty effects*

Effect	Estimated Magnitude	Justification
Trace gas absorption (CO, CH, NO)	<0.001	Window channel placement
Rayleigh scattering altitude dependence	<0.002	Accurate DEM + pressure correction
LUT interpolation errors	<0.001	Dense sampling validated vs. 6S
Aerosol hygroscopic growth	<0.005	Implicit in fixed RH=80% model
Polarization	<0.003	Minor for broad-band sensors
Spectral band mismatch	<0.002	SMAC computed for actual RSR

Combined second-order: $u(0) \approx 0.007$ (added quadratically to total, configurable via [Coefficients] u2_0 in config).

5.4 Uncertainty Characterization Summary

5.4.1 Input Uncertainty Estimates

Table 17: *Input uncertainty estimates summary*

Input Parameter	Source	Uncertainty Estimate	Notes
R_{toa}	Level 1C FDR	From DTOA-r	Band-dependent, separated into random/structured/common
U_{O_3}	MERRA-2 + literature	6%	Wargan et al. (2017), Davis et al. (2017)
U_{H_2O}	MERRA-2 + estimate	20%	Conservative, needs validation
P_0	MERRA-2 SLP	<1 hPa	Meteorological model uncertainty
λ (lapse rate)	Model assumption	0.002 K/m	Van Besselaar et al. (2011)
z (elevation)	GTOPO30 + geoloc	Δ_{elev} (DEM stddev)	Convolution with $\sigma = 0.5$ pixels (~ 0.3 VGT pixel)
τ_a^{550}	MERRA-2 + validation	$\max(0.03, 0.15E\tau_a)$	Gueymard & Yang (2020), regionalization planned
Aerosol model	AMIP ensemble	Ensemble spread	10-member AMIP runs

Input Parameter	Source	Uncertainty Estimate	Notes
RTM approximations	Modelling	1-30%	Scene dependent. Worst case: rough terrain, high AOD

5.4.2 Correlation Structures

Table 18: *Correlation structures for propagated uncertainty effects*

Effect	Spatial Correlation	Temporal Correlation	Implementation
TOA-r (random)	None	None	Average as $\sqrt{\Sigma u^2}/N$
TOA-r (structured)	Scan-line or column	None	Use L1C correlation matrix
TOA-r (common)	Global	Global	Simple average (no reduction)
U_{O_3}	~500 km (synoptic)	~3 days	Exponential decay
U_{H_2O}	~200 km (convective)	~1 day	Exponential decay
P_s (meteorological)	Global	>1 month	Common (no reduction)
P_s (elevation)	DEM resolution	Permanent	Structured (terrain-correlated)
τ_a^{550}	~200-500 km (plume)	~1 week	Exponential decay
Aerosol model	1ř (climatology grid)	1 week	Rectangle function
RT approximations	Scene-dependent	None	Random or systematic (case-by-case)

5.5 Limitations and Future Improvements

5.5.1 Known Limitations

Table 19: *Known limitations and future improvements*

Limitation	Impact	Status
Surface BRDF: Simplified kernel model	Up to 10% error in directional reflectance over non-forest surfaces	Documented
Terrain effect: Simplified model	Several % in mountainous regions	Documented
Water Vapor Uncertainty: Conservative 20% estimate	May overestimate total uncertainty in dry conditions	Future: Refined characterization from MERRA-2 validation
Adjacency Effects: Not corrected	Significant for high-contrast scenes (coast, cloud edge)	Known limitation — no correction applied
Köppen-Geiger regionalization	Regional AOD uncertainty multipliers not applied	Planned feature — data and multipliers defined but not implemented

6. Conclusions

This ATBD demonstrates a comprehensive implementation of the FIDUCEO uncertainty propagation methodology for atmospheric correction of satellite vegetation sensors. Key achievements:

- **Rigorous Uncertainty Quantification:** All major uncertainty sources are identified, characterized, and propagated through the measurement equation
- **Operational AOD Uncertainty Model:** Based on extensive MERRA-2 validation (Gueymard & Yang 2020), providing realistic AOD uncertainties: $\sigma(\tau_a) = \max(0.03, 0.15 \times \tau_a)$ with regional adjustments planned
- **Ensemble Approach for Model Uncertainty:** AMIP 10-member ensemble quantifies aerosol optical property uncertainties
- **Efficient GPU Implementation:** SMAC-CL enables fast processing while computing full Jacobian matrices
- **Validation Framework:** Following Aerosol_cci and FIDUCEO best practices (Popp & Mittaz 2022)

This approach provides climate-quality TOC reflectances with fully traceable, metrologically sound uncertainties suitable for:

- Long-term surface monitoring
- Multi-sensor harmonization
- Climate model validation

7. References

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8. Appendix A: SMAC Coefficient Calculation

8.1 Spectral Response Functions and Gaseous Absorption

Radiative transfer computations are done for all camera RSRFs (CENTER, LEFT, RIGHT). Typical atmospheric gaseous transmissions are computed for each Proba-V spectral band.

8.2 Processing Chain

SMAC coefficients are derived from basic data through:

1. **Spectroscopic data** (HITRAN 2012/2016) Py4Cats/HAPI line-by-line radiative transfer
2. **Gaseous absorption regression** SMAC coefficients for O and HO transmission fits
3. **ARTDECO RT simulations** TOA reflectance for dense geometry/pressure/AOD sampling
4. **Multi-dimensional polynomial fitting** SMAC scattering coefficients
5. **Fit accuracy validation** RMSE vs 6S, neural network verification

8.3 Gaseous Absorption Modelling

With the RSRF data, typical absorption is computed based on the HITRAN 2012/2016 database and Py4Cats/HAPI. Regression of gaseous transmission versus gas concentrations, air masses, and barometric pressure yields SMAC coefficients:

$$T = \exp(a \cdot (m \cdot U_{H_2O})^n)$$

Where m is the air mass and U_{H_2O} is the water vapor column.

8.4 Radiative Transfer Computations and Fit Accuracy

Radiative transfer equation for each pixel (geometry, band, AOT, P_{surf}) is solved using ARTDECO RT package. TOA reflectance computations are done for a black surface and no gaseous absorption:

- **RAA:** 121 values from 0 to 180°

- **SZA:** 38 values from 0 to 89° (used only if <70°)
- **VZA:** 47 values from 0 to 89° (used only if <70°)
- **Surface pressure:** 6 values from 600 to 1050 hPa
- **AOT at 550 nm:** 16 values from 0.01 to 1.5 (used only if < 0.8)

Several multi-dimensional fits are performed to give the SMAC model parameters. Fit accuracy is validated against exact ARTDECO computations.

8.5 Fit Uncertainty Model (ONNX)

A lightweight Feed-Forward Neural Network (MLP) in ONNX format estimates SMAC fit uncertainty.

Inputs: Angles (SZA, VZA, RAA), AOD at 550 nm, pressure, TOC-r, iaer, TOA-r, wavelength

Performance: Mean Square Error of 0.0097 on predicted RTM uncertainty on R_{toc} .

Runtime: `onnxruntime.InferenceSession` loaded once per worker process.

8.6 BRDF Effect Estimation

The baseline AC is run under the assumption of a Lambertian surface. If the surface BRDF shape is known, the RossThick-LiSparse-R model is used:

$$\rho_g(\theta_s, \theta_v, \phi) = k_0 + k_1 F_1(\theta_s, \theta_v, \phi) + k_2 F_2(\theta_s, \theta_v, \phi) \quad (25)$$

The fast RTM with BRDF uses the 6S formulation with modified total atmospheric transmission $T_{atm,g}$ (Ramon and Santer, 2001). The 12 regression coefficients d_i^k and b_i^k for kernel directional averages are stored in the SMAC coefficient structure.

9. Appendix B: Aerosol Model Definitions

9.1 Aerosol Variability on Earth

44,688 MERRA-2 grid points over land. Four dates represent seasonal variability: 2011/01/01, 2011/04/01, 2011/07/01, and 2011/10/01 at 13:00.

Aerosol models are defined according to MERRA-2 5 AOT ratios:

$$X_u = \frac{\tau_{a;u}^{550}}{\tau_a^{550}}, \quad u \in \{DU, SU, OC, BC, SS\}$$

Each mixing ratio X_u covers 0-100% range (except $BC < 25\%$). From 5000 possible combinations, 148 combinations effectively occur on the 44,688 pixels of 2011/07/01.

9.2 Aerosol Optical Properties

MERRA-2 aerosol components are mapped to OPAC models (RH=80%):

Table 20: *MERRA-2 to OPAC aerosol component mapping*

MERRA-2 Component	OPAC Model/Component
SU/SU	Antarctic model
DU/DU	Desert model
OC/OM	Water soluble (waso)
SS/SS	Maritime clean
BC/BC	Soot

Spectral dependence: $K(\lambda) = \sum_u K_u(\lambda) X_u$

Single scattering albedo: $\varpi_0(\lambda) = \sum_u \varpi_0^u(\lambda) X_u$

Phase function: $P(\theta, \lambda) = \sum_u P_u(\theta, \lambda) X_u$

Asymmetry parameter: $g(\lambda) = \frac{1}{2} \int_{-1}^1 \mu P(\mu, \lambda) d\mu$

10. Appendix C: SMAC-CL Implementation

10.1 Input Reading

For each pixel, the following quantities are read from Level 1 files:

- TOA radiance: TOA-r()
- Error: DTOA-r()
- Solar extra-terrestrial irradiance: Es()
- Geometry: SZA, SAA, VZA, VAA, VZA_SWIR, VAA_SWIR
- Location: lat, lon

Sensor name, date, and mean acquisition time (UTC) are stored as file attributes for SMAC coefficient and MERRA-2 file selection.

10.2 Pixel Masking

Cloud-contaminated pixels are masked (`clm != 0`). Pixels with $SZA > 90^\circ$ are also excluded.

10.3 Spatial and Temporal Interpolation

- **DEM:** Linear interpolation (improvement over nearest-neighbor in v1.0)
- **MERRA-2:** `RegularGridInterpolator(method='linear')` — bilinear spatial + linear temporal
- **AMIP:** Spatial interpolation only (monthly climatology, no temporal interpolation)

Unit conversions:

Table 21: *Unit conversions applied in SMAC-CL*

Variable	MERRA-2 Unit	SMAC Unit	Conversion
U_{O_3}	Dobson	cm.atm	1e-3
U_{H_2O}	kg.mš	g.cmš	1e-1
P_0	Pa	hPa	1e-2

Surface pressure P_s and uncertainty ΔP_s are computed using Eqs. (21)-(24).

10.4 Aerosol Model Selection

$$iaero = \operatorname{argmin}_i \sum_u (X_u - X_{u,i})^2, \quad i = 0..147$$

The $X_{u,i}$ basis for 148 predefined models is stored in `Aerosol_model_fraction.txt` and read once.

10.5 SMAC-CL Function

Architecture:

- Language: OpenCL (compatible with GPU and CPU)

- Interface: Python wrapper (smaccl library)
- Processing mode: Pixel-parallel (massively parallel on GPU)

Main Processing Function:

```
def smac_cl_run(
    coeffs,          # SMAC coefficients (148 aerosol models)
    tetas, tetav,    # SZA, VZA arrays
    phis, phiv,      # SAA, VAA arrays
    uh2o, uo3,       # Water vapor, ozone columns
    taup550,         # AOD at 550 nm
    pressure,        # Surface pressure
    rtoa,            # TOA reflectance
    k1p, k2p,        # BRDF kernel ratios (default 0 for Lambertian)
    iaero_list       # List of aerosol model indices (11 models)
):
    """
    Inputs: All arrays dimensioned [npixels] or [nbands, npixels]
    Outputs:
        - rtoc: TOC reflectance [nbands, npixels]
        - jacobians: Sensitivity matrices
        - u_ensemble: Std dev from aerosol ensemble
    """
```

Processing Flow:

1. Reorganize input arrays for GPU efficiency (transpose, flatten, pad)
2. Upload data to GPU/CPU memory
 - TOA reflectances, geometry, auxiliary data
 - SMAC coefficients (pre-loaded once)
3. GPU/CPU kernel execution (per-pixel parallel)
 - a. Compute gaseous transmissions
 - b. Compute Rayleigh + aerosol scattering
 - c. Apply SMAC equations -> Rtoc
 - d. Compute analytical Jacobians
 - e. Finite-difference Jacobians (Ps, tau550)
 - f. 10 AMIP ensemble members
4. Download results from GPU/CPU (Rtoc, Jacobians, ensemble statistics)

5. CPU post-processing

- Combine uncertainty components
- Apply terrain corrections
- Quality flag assignment

Performance:

- GPU acceleration: ~100x faster than CPU for large images
- Memory requirement: ~2 GB GPU RAM for full-swath processing
- Typical processing time: <1 second per tile (1000x1000 pixels)

10.6 Level 2A Output Variables

Table 22: *Level 2A standard output variables*

Output	Description	Dimensions	Dtype
rTOC	TOC directional reflectance	(bands, height, width)	float32
UrTOC	Total uncertainty	(bands, height, width)	float32
Jrtoa	Jacobian vs Rtoa	(bands, height, width)	float32
Juo3	Jacobian vs UO3	(bands, height, width)	float32
Juh2o	Jacobian vs UH2O	(bands, height, width)	float32
Jpre	Jacobian vs Ps	(bands, height, width)	float32
Jtau550	Jacobian vs AOD	(bands, height, width)	float32
flag	Quality flags	(height, width)	int16
iaero	Aerosol model index	(height, width)	int32
TOTEXTTAU	AOD 550 used	(height, width)	float32
SZA, SAA, VZA, VAA	Geometry	(height, width)	float32
clm	Cloud mask	(height, width)	uint8
SM_MAP_B1..B4	Status maps	(height, width)	uint8

Diagnostic outputs (additional):

Table 23: *Level 2A diagnostic output variables*

Output	Description
unc_h2o	HO uncertainty component
unc_o3	O uncertainty component
unc_ps	Pressure uncertainty component
unc_aot	AOD uncertainty component
slope_err	Terrain slope error
urtoc_terrain	Terrain uncertainty contribution
UrTOC_rtm_brdf	BRDF uncertainty contribution
UrTOC_rtm_fit	RTM fit uncertainty contribution

11. Appendix D: Implementation Notes

This appendix documents implementation-specific details, performance optimizations, and known limitations of the FDR4VGT production codebase. These are not changes to the scientific algorithm but rather engineering decisions for operational efficiency.

11.1 Performance Optimizations

11.1.1 Ancillary Zarr Cache

Table 24: *Ancillary Zarr cache optimization*

Aspect	Description
What	Pre-interpolates MERRA2 fields to full satellite grid once, stores in local Zarr cache
Why	MERRA2 source files on network FS; per-tile interpolation re-reads network and repeats work
Function	<code>precompute_ancillary()</code> in <code>process.py</code>
Toggle	[Sizes] <code>anc_cache</code> (default on)
Numerical equivalence	Pointwise interpolation identical to per-tile interpolation (halo reads from cache)
Memory	Streamed tile-by-tile to bound peak memory (~2.8 GB safe on 4 GB nodes)

Cache format: Zarr format 2, most fields as float16 (small values, sufficient precision), SLP and T10M as float32 (overflow protection).

11.1.2 Tile-Based Processing

Table 25: *Tile-based processing optimization*

Aspect	Description
What	Processes image in overlapping tiles (<code>batch_size</code> & <code>batch_size</code> with halo)
Why	Bounded memory per tile, parallel processing with workers
Function	<code>process_batched()</code> generates tile grid, <code>_worker_compute_tile()</code> processes each
Config	[Sizes] <code>chunks_size</code> (default 512), [Sizes] <code>nworkers</code> (default 1)
Halo	1 pixel halo for terrain slope/aspect computation (edge effects)

11.1.3 Local Output File

Table 26: *Local output file optimization*

Aspect	Description
What	Writes output NetCDF to local temp file, moves to final destination once at end
Why	Serial per-tile append writes degrade on network FS (ceph) — 2s 5s/tile
Function	<code>process_batched()</code> creates temp file, <code>shutil.move()</code> at end
Toggle	[Sizes] <code>local_output</code> (default on)

11.1.4 Single NetCDF Handle

Table 27: *Single NetCDF handle optimization*

Aspect	Description
What	Opens output NetCDF handle once, reuses for all tiles
Why	Per-tile open/close/flush was dominant writer cost (HDF5 overhead)
Function	<code>nc_out = Dataset(output_path, 'a')</code> reused across tiles
Flush	Periodic close+reopen every <code>flush_interval</code> tiles to free HDF5 cache

11.1.5 DEM Cropping

Table 28: *DEM cropping optimization*

Aspect	Description
What	Crops global GTOPO30 DEM to product's lat/lon extent before loading
Why	Full DEM ~1.9 GB, cropped ~0.2 GB (narrow lon range)
Function	<code>xrcrop(dem_ds, lat=data['y'], lon=data['x'])</code>

11.1.6 Copy-on-Write Shared State

Table 29: *Copy-on-write shared state optimization*

Aspect	Description
What	Pre-forks worker pool, then populates <code>_SHARED</code> dict (inherited via COW)
Why	Avoids pickling large objects (SMAC coefficients, DEM, MERRA2) per task
Function	<code>_SHARED.update()</code> before pool creation
Objects	SMAC coefficients (~100 MB), aerosol model LUT, terrain data

11.2 Data Type Requirements

11.2.1 Float32 for SLP/T10M (Overflow Fix)

Table 30: *Float32 overflow fix for SLP/T10M*

Aspect	Description
Issue	Ancillary Zarr cache stored T10M and SLP as float16
Root cause	<code>smaccl.c3s_lib.Ps()</code> computes $\log(R*T)$ where $R*T = 287.058 \times 290 = 83246$
Overflow	83246 exceeds float16 max (65504) NaN in pressure correction NaN rTOC
Fix	<code>_MERRA_CACHE_FLOAT32_VARS = {'SLP', 'T10M'}</code> keeps these fields as float32
Commit	51f7e68 — “Fix ancillary cache dtype overflow causing NaN rTOC”

Validation: Full processing run (100 tiles, 241.9s) shows rTOC finite on land ~77.2%, ocean ~0% (expected — ocean pixels masked by cloud/quality flags).

11.2.2 Float16 for Small-Value Fields

Fields with small values (~0-0.1 range) are safely stored as float16:

- `ERROR` (uncertainty components)
- `TOTEXTTAU` (AOD values, typically <1.0)
- Species fractions (0-1 range)

These are widened to float32 on NetCDF write (netCDF4 has no native half-float type).

11.3 Known Limitations

11.3.1 Adjacency Effects

Status: Not corrected.

Impact: Significant for high-contrast scenes (coast, cloud edge, urban/rural boundaries). The SMAC algorithm assumes a horizontally homogeneous scene within the pixel footprint. Adjacent

pixels with different reflectances contribute to the measured TOA radiance through atmospheric scattering, which is not accounted for.

Future work: Implement semi-empirical adjacency correction (e.g., based on scene contrast and atmospheric conditions).

11.3.2 Köppen-Geiger Regionalization

Status: Planned feature.

Description: The AOD uncertainty model (Eq. 23) uses global multipliers. Regional multipliers based on Köppen-Geiger climate classification are defined but not yet implemented (see Table 14).

Data available: `kg_zoneNumGueymard.csv` in `ANCILLARY` directory.

11.3.3 Water Vapor Uncertainty

Status: Conservative 20% estimate.

Impact: May overestimate total uncertainty in dry conditions.

Future work: Refined characterization from MERRA-2 validation studies.

11.4 Tile Processing Architecture

11.4.1 Overview

The processing pipeline uses a tile-based architecture for memory-bounded, parallel execution:

`process_batched()` -- Main entry point

1. Generate tile grid from input extent
 - Tile size: `chunks_size` x `chunks_size`
 - Overlap: +/-1 pixel halo
 - Total tiles: `ceil(y/chunks)` x `ceil(x/chunks)`
2. Precompute ancillary cache (if enabled)
 - Interpolate MERRA2 to full grid
 - Save to local Zarr cache
3. Create worker pool (`nworkers`)
 - Fork workers before populating `_SHARED`
 - COW inheritance of large objects
4. Process tiles in parallel
 - `_worker_compute_tile()` per tile

- SMAC-CL run (VNIR + SWIR)
 - Uncertainty propagation
 - Write tile to NetCDF
5. Move local output to final destination

11.4.2 Worker Function

`_worker_compute_tile()` performs end-to-end tile computation:

1. **Read tile data** from input L1C and ancillary cache
2. **Interpolate DEM** and compute terrain (slope/aspect)
3. **Select aerosol model** (closest in 5D species space)
4. **Run SMAC-CL** (VNIR bands, then SWIR band)
5. **Compute BRDF effect** (Lambertian vs BRDF run, if enabled)
6. **Compute terrain effect** (CESBIO model)
7. **Compute RTM fit uncertainty** (ONNX model)
8. **Propagate uncertainties** (`compute_urtoe()`)
9. **Build quality flags** (`build_flag()`)
10. **Write tile** to output NetCDF

11.5 Output Format Details

11.5.1 NetCDF Structure

The output NetCDF file follows CF-1.8 conventions with attributes from `cf_attributes_res2.json`.

Dimensions:

Table 31: *Output NetCDF dimensions*

Dimension	Size	Description
<code>bands</code>	4	B1 (Blue), B2 (Red), B3 (NIR), B4 (SWIR)
<code>height</code>	Variable	Tile height (y dimension)
<code>width</code>	Variable	Tile width (x dimension)

Global attributes:

Table 32: *Output NetCDF global attributes*

Attribute	Description
<code>sensor</code>	Sensor name (Proba-V, SPOT-VGT)
<code>date</code>	Acquisition date
<code>time</code>	Mean acquisition time (UTC)
<code>projection</code>	Projection attributes from input L1C
<code>conventions</code>	CF-1.8
<code>history</code>	Processing timestamp and version

Variable attributes: Each variable has `standard_name`, `units`, `long_name`, and `comments` attributes applied from the CF JSON configuration.

11.5.2 Chunk Cache Optimization

The output NetCDF is written with optimized chunk sizes matching the tile processing grid, ensuring efficient random access for downstream consumers.

12. Appendix E: SMACCL OpenCL Kernel API

This appendix documents the SMACCL OpenCL function API as implemented in the HYGEOS `smaccl` repository (OpenCL branch), including both:

- Python host wrapper API (`smaccl/smaccl.py`, class `Smaccl`)
- OpenCL kernel API (`smaccl/devicecl.cl`, kernel `smaccl`)

The objective is to provide a precise implementation-facing contract for integration and maintenance.

12.1 Python Host API

12.1.1 Signature

```
def run(  
    self,  
    coeffs,  
    tetas,  
    tetav,  
    phis,  
    phiv,  
    uh2o,  
    uo3,  
    taup550,  
    pression,  
    rtoa,  
    k1p,  
    k2p,  
    iaero,  
    NBLOOP=1,  
)
```

12.1.2 Returned tuple

```
(rsurf, rsurf_0, dev_std, Jrtoa, Juo3, Juh2o, Jpre, Jtaup)
```

All returned arrays are `float32` and shaped like `rtoa`.

12.1.3 Input semantics

- `coeffs`: SMAC coefficients. First dimension defines number of spectral bands processed.
- `tetas`: solar zenith angle (SZA), `float32`.
- `tetav`: view zenith angle (VZA), `float32`.
- `phis`: solar azimuth angle (SAA), `float32`.
- `phiv`: view azimuth angle (VAA), `float32`.
- `uh2o`: total water vapor column, `float32`.
- `uo3`: total ozone column, `float32`.
- `taup550`: aerosol optical depth at 550 nm, `float32`.
- `pression`: surface pressure, `float32`.
- `rtoa`: TOA reflectance, `float32`.
- `k1p`, `k2p`: BRDF normalized kernel ratios. Set to 0 for Lambertian processing.

- **iaero**: aerosol model indices per pixel, with **Naero** candidates.
- **NBLOOP**: number of internal per-thread loops (used for repeated runs/MC patterns).

12.1.4 Shape contract

The wrapper supports 4D tensor layout for tiled processing:

- **rtoa**: (NBAND, NZ, XBLOCK, XGRID)
- Angular and ancillary scalar fields (**tetas**, **tetav**, **phis**, **phiv**, **uh2o**, **uo3**, **taup550**, **pression**): (NZ, XBLOCK, XGRID)
- BRDF fields (**k1p**, **k2p**): (NBAND, NZ, XBLOCK, XGRID)
- Aerosol index tensor (**iaero**): (Naero, NZ, XBLOCK, XGRID)

Where:

- **NBAND**: number of spectral bands handled in current call
- **NZ**: padded depth along pixel index dimension
- **XBLOCK**, **XGRID**: OpenCL launch decomposition
- **Naero**: number of aerosol candidates used in ensemble/selection

12.1.5 Runtime behavior

1. Allocate output buffers for **rsurf**, **rsurf_0**, **dev_std**, Jacobians.
2. Copy host arrays to OpenCL buffers.
3. Launch OpenCL kernel **smaccl**.
4. Copy outputs from device to host arrays.

12.2 OpenCL Kernel API

12.2.1 Signature

```
__kernel void smaccl(  
    __global coef_atmos *ca,  
    __global float *tetas_,  
    __global float *tetav_,  
    __global float *phis_,  
    __global float *phiv_,  
    __global float *uh2o_,
```

```
__global float *uo3_,
__global float *taup550_,
__global float *pression_,
__global float *rtoa_,
__global float *rsurf,
__global float *Jr,
__global float *Juo3,
__global float *Juh2o,
__global float *Jpre,
__global float *Jtaup,
__global float *k1p_,
__global float *k2p_,
__global int *iaero,
int NMOD,
int NBLOOPd,
int NBANDd,
int NZd
)
```

12.2.2 Kernel-level constants and finite differences

The implementation uses:

- `dpre` = 10.0 hPa for pressure Jacobian finite difference
- `dtau_rel` = 0.1 for relative AOD perturbation
- `NRUN` = 3 + `NBLOOPd`

Interpretation of `NRUN`:

- 1 reference run
- 2 perturbation runs for finite-difference Jacobians (`P_s`, `tau550`)
- `NBLOOPd` optional additional looped runs

12.3 Output Mapping

The host API output tuple maps to AC physics as:

Table 33: *SMACCL kernel output variable mapping*

Host variable	Physical meaning
<code>rsurf</code>	TOC directional reflectance (<code>rTOC</code>)
<code>rsurf_0</code>	Lambertian-equivalent TOC reflectance (<code>rTOC_0</code>)
<code>dev_std</code>	Aerosol-model ensemble spread (<code>UrTOC_ens</code>)
<code>Jrtoa</code>	Jacobian $\partial R_{toc}/\partial R_{toa}$
<code>Juo3</code>	Jacobian $\partial R_{toc}/\partial U_{O3}$
<code>Juh2o</code>	Jacobian $\partial R_{toc}/\partial U_{H2O}$
<code>Jpre</code>	Jacobian $\partial R_{toc}/\partial P_s$
<code>Jtaup</code>	Jacobian $\partial R_{toc}/\partial \tau_{550}$

12.4 ISmaccl Wrapper API

The integration layer `ISmaccl` wraps the kernel API for product-level execution.

12.4.1 Constructor

```
ISmaccl(
    config,
    frac_aer_model,
    ca,
    XBLOCK=128,
    XGRID=128,
    breakpoint=False,
    ancillary=False,
    platform=..., # forwarded to Smaccl(platform=...)
)
```

12.4.2 Product-level methods

- `run_block(l2_data, batch_size=512)`: block mapping over xarray/dask chunks.
- `run(l2_data, iaero)`: scene/tile execution and assembly of xarray output dataset.
- `exec_smaccl(...)`: low-level preparation/padding/reshaping, kernel call, post-mapping.

12.4.3 `exec_smaccl` returned tuple

```
(  
    rsurf,  
    rsurf0,  
    dev_std,  
    Juh2o,  
    Juo3,  
    Jrtoa,  
    Jpre,  
    Jtaup,  
    Duh2o,  
    Duo3,  
    Drtoa,  
    Dpre,  
    Dtaup,  
)
```

This extends kernel outputs by including uncertainty-input terms (D^*) computed on host side for subsequent propagation.

12.4.4 Band handling in Proba-V mode

For Proba-V processing, the wrapper executes two calls:

1. VIS/NIR group using VZA/VAA
2. SWIR group using VZA_IR/VAA_IR

Then it merges outputs into full 4-band products.

12.5 Device and Data Contracts

12.5.1 Data type and ordering requirements

- Arrays converted to C-order contiguous layout
- Floating inputs cast to `float32`
- Aerosol index array cast to integer (`int16/int32` depending stage)
- Invalid geolocation values sanitized before kernel packing
- Padded extended arrays (`GSIZEXT`) filled with NaN then reshaped to launch geometry

12.5.2 Device and queue selection

`Smaccl.__init__` compiles kernel source and creates queue/context via `set_queue(xpu, environment)`.

Supported environment mapping in repository implementation includes:

Table 34: *OpenCL device/environment mapping*

Environment	CPU device name	GPU device name
Default	Portable Computing Language	NVIDIA CUDA
Turenne	Intel(R) OpenCL	—
TerraScope	Intel(R) OpenCL	NVIDIA CUDA

If no compatible/available device is found, initialization fails.

12.5.3 Contractual outputs for FDR4VGT

For this ATBD implementation, the contractual kernel-facing outputs are:

- Reflectance: `rTOC`, `rTOC_0`
- Jacobians: `Jrtoa`, `Juo3`, `Juh2o`, `Jpre`, `Jtau550`
- Aerosol spread: `UrTOC_ens`

Host-side uncertainty terms (`Duh2o`, `Duo3`, `Drtoa`, `Dpre`, `Dtaup`) are combined with Jacobians in the uncertainty propagation equation documented in the Uncertainty Analysis section.

This appendix defines the implementation API used by the operational chain and should be updated if the upstream HYGEOS `smaccl` API changes.